

Table 1: Performance (%) across 25 node classification benchmarks comparing 14 supervised methods and 7 graph representation learning methods. All baselines use *fixed hyperparameter setting* (Fixed-HP), where a single hyperparameter configuration is selected based on the best average validation accuracy across all datasets and then applied uniformly to every dataset; models are still trained separately on each dataset. In contrast, Node4All is trained solely on synthetic graphs and applied uniformly across all benchmarks without any per-dataset training or tuning. The **best and **second-best** methods in terms of average accuracy are highlighted. Despite never observing target datasets during training, Node4All achieves the best overall performance, demonstrating its effectiveness in generalizing to arbitrary graph datasets within a single model.**

Methods	Actor	Amazon Computer	Amazon Photo	Amazon Ratings	Blog Catalog	Chameleon	Citeseer	Coauthor CS	Coauthor Physics	Cora	Cornell	DBLP	Deezer
Fixed Hyperparameter Setting													
Linear	34.35 $\pm$ 0.96	69.75 $\pm$ 1.29	77.77 $\pm$ 1.17	38.61 $\pm$ 0.44	68.97 $\pm$ 1.34	34.10 $\pm$ 2.53	49.05 $\pm$ 0.74	84.28 $\pm$ 0.58	84.24 $\pm$ 1.78	47.44 $\pm$ 0.68	69.73 $\pm$ 7.33	55.28 $\pm$ 0.76	55.20 $\pm$ 2.38
MLP	34.79 $\pm$ 0.98	69.66 $\pm$ 0.78	78.41 $\pm$ 1.49	45.77 $\pm$ 0.38	73.53 $\pm$ 0.78	39.75 $\pm$ 4.63	51.60 $\pm$ 1.39	87.01 $\pm$ 0.62	88.02 $\pm$ 1.28	55.79 $\pm$ 1.14	72.70 $\pm$ 5.47	55.57 $\pm$ 0.74	57.16 $\pm$ 2.10
GIN	24.75 $\pm$ 4.50	37.05 $\pm$ 7.61	23.82 $\pm$ 5.11	44.86 $\pm$ 3.26	18.21 $\pm$ 2.49	29.63 $\pm$ 6.34	53.57 $\pm$ 4.91	84.50 $\pm$ 0.58	77.08 $\pm$ 21.00	70.26 $\pm$ 1.86	36.49 $\pm$ 7.67	68.47 $\pm$ 2.56	55.22 $\pm$ 2.08
GAT	28.54 $\pm$ 1.00	83.59 $\pm$ 1.28	90.44 $\pm$ 1.38	49.09 $\pm$ 0.61	48.13 $\pm$ 9.74	39.97 $\pm$ 3.25	67.81 $\pm$ 1.25	89.86 $\pm$ 0.55	91.81 $\pm$ 0.90	79.58 $\pm$ 1.39	41.89 $\pm$ 5.02	72.84 $\pm$ 2.40	54.51 $\pm$ 0.74
GAT2	27.51 $\pm$ 0.75	84.78 $\pm$ 0.90	89.86 $\pm$ 1.21	49.86 $\pm$ 0.55	49.33 $\pm$ 12.48	39.63 $\pm$ 2.82	67.57 $\pm$ 1.38	89.00 $\pm$ 0.67	92.03 $\pm$ 0.82	78.94 $\pm$ 1.37	40.81 $\pm$ 4.75	74.76 $\pm$ 2.65	55.85 $\pm$ 0.38
SGC	29.27 $\pm$ 1.15	85.40 $\pm$ 1.06	91.79 $\pm$ 0.46	42.19 $\pm$ 0.50	71.38 $\pm$ 1.04	41.59 $\pm$ 4.07	67.18 $\pm$ 0.40	90.34 $\pm$ 0.54	92.73 $\pm$ 0.64	78.22 $\pm$ 0.47	43.24 $\pm$ 7.93	67.39 $\pm$ 3.49	55.05 $\pm$ 1.51
S ² GC	29.34 $\pm$ 1.00	85.51 $\pm$ 1.10	91.95 $\pm$ 0.41	42.15 $\pm$ 0.38	78.74 $\pm$ 1.40	39.46 $\pm$ 3.42	66.71 $\pm$ 0.44	90.65 $\pm$ 0.48	91.74 $\pm$ 0.90	78.21 $\pm$ 0.28	40.54 $\pm$ 5.41	69.98 $\pm$ 3.33	55.02 $\pm$ 1.68
JKNet	29.74 $\pm$ 1.14	85.37 $\pm$ 0.84	91.86 $\pm$ 0.55	48.85 $\pm$ 0.83	71.06 $\pm$ 1.24	39.38 $\pm$ 3.36	69.38 $\pm$ 0.71	90.63 $\pm$ 0.63	93.12 $\pm$ 0.43	81.04 $\pm$ 0.85	43.51 $\pm$ 5.85	74.11 $\pm$ 3.57	56.00 $\pm$ 0.84
APPNP	31.28 $\pm$ 0.94	85.27 $\pm$ 1.14	92.03 $\pm$ 0.42	49.28 $\pm$ 0.60	86.11 $\pm$ 2.50	38.88 $\pm$ 2.92	68.56 $\pm$ 0.41	90.73 $\pm$ 0.61	93.64 $\pm$ 0.43	81.50 $\pm$ 0.54	38.65 $\pm$ 4.99	75.32 $\pm$ 3.14	56.20 $\pm$ 1.09
GCN	29.41 $\pm$ 0.75	85.36 $\pm$ 0.90	91.59 $\pm$ 0.49	49.53 $\pm$ 0.37	70.46 $\pm$ 1.27	39.29 $\pm$ 2.68	68.67 $\pm$ 0.69	90.34 $\pm$ 0.69	93.43 $\pm$ 0.49	81.07 $\pm$ 0.36	40.81 $\pm$ 5.98	70.88 $\pm$ 3.20	55.50 $\pm$ 1.76
GCN2	33.7 $\pm$ 1.06	84.65 $\pm$ 0.78	91.37 $\pm$ 0.52	49.39 $\pm$ 0.78	84.39 $\pm$ 2.40	40.46 $\pm$ 3.67	69.00 $\pm$ 1.35	89.00 $\pm$ 0.52	92.90 $\pm$ 0.52	80.55 $\pm$ 1.33	41.35 $\pm$ 3.64	73.95 $\pm$ 3.31	57.02 $\pm$ 0.98
SAGE	34.31 $\pm$ 0.72	83.83 $\pm$ 0.66	91.33 $\pm$ 0.69	50.48 $\pm$ 0.57	82.30 $\pm$ 1.82	41.04 $\pm$ 3.91	68.42 $\pm$ 0.65	90.34 $\pm$ 0.57	93.04 $\pm$ 0.55	80.14 $\pm$ 0.76	53.51 $\pm$ 5.10	71.54 $\pm$ 2.46	56.35 $\pm$ 0.94
GPR-GNN	34.06 $\pm$ 0.69	86.61 $\pm$ 0.80	92.98 $\pm$ 0.55	48.75 $\pm$ 0.65	88.53 $\pm$ 1.59	38.81 $\pm$ 2.23	69.83 $\pm$ 0.91	91.45 $\pm$ 0.80	93.68 $\pm$ 0.58	81.25 $\pm$ 0.78	56.49 $\pm$ 6.22	72.05 $\pm$ 2.71	56.40 $\pm$ 1.10
UniMP	34.86 $\pm$ 0.83	83.80 $\pm$ 0.83	91.55 $\pm$ 0.85	51.16 $\pm$ 0.80	73.36 $\pm$ 1.68	42.02 $\pm$ 3.49	69.09 $\pm$ 0.54	90.49 $\pm$ 0.64	92.38 $\pm$ 0.63	80.00 $\pm$ 0.73	54.59 $\pm$ 5.10	72.25 $\pm$ 2.35	56.29 $\pm$ 1.05
node2vec	34.14 $\pm$ 1.52	78.76 $\pm$ 0.86	87.29 $\pm$ 1.10	50.98 $\pm$ 0.31	78.81 $\pm$ 1.30	40.45 $\pm$ 2.49	62.06 $\pm$ 0.56	88.95 $\pm$ 0.88	88.71 $\pm$ 1.32	73.49 $\pm$ 0.70	69.73 $\pm$ 4.95	68.55 $\pm$ 1.28	55.32 $\pm$ 1.85
GAE	29.66 $\pm$ 1.28	83.85 $\pm$ 0.75	91.08 $\pm$ 0.61	49.59 $\pm$ 0.83	72.76 $\pm$ 1.14	43.87 $\pm$ 2.86	59.36 $\pm$ 0.71	91.25 $\pm$ 0.40	92.92 $\pm$ 0.56	79.59 $\pm$ 1.00	41.62 $\pm$ 5.69	80.70 $\pm$ 1.11	54.55 $\pm$ 1.14
BGRL	29.81 $\pm$ 1.11	83.28 $\pm$ 0.80	90.23 $\pm$ 0.84	50.99 $\pm$ 0.41	74.74 $\pm$ 0.82	43.17 $\pm$ 3.63	66.09 $\pm$ 1.12	91.21 $\pm$ 0.47	93.76 $\pm$ 0.50	82.25 $\pm$ 0.45	39.19 $\pm$ 5.30	80.22 $\pm$ 1.21	54.53 $\pm$ 1.60
CCA-SSG	29.88 $\pm$ 0.79	85.12 $\pm$ 0.50	90.84 $\pm$ 0.90	49.78 $\pm$ 0.62	73.92 $\pm$ 1.18	44.89 $\pm$ 3.14	68.76 $\pm$ 1.25	90.67 $\pm$ 0.50	93.83 $\pm$ 0.37	83.22 $\pm$ 0.30	45.14 $\pm$ 8.55	81.75 $\pm$ 0.81	56.23 $\pm$ 1.70
GraphMAE	29.78 $\pm$ 0.72	83.48 $\pm$ 0.92	90.74 $\pm$ 0.63	49.92 $\pm$ 0.82	62.77 $\pm$ 2.16	40.17 $\pm$ 3.39	70.05 $\pm$ 0.48	91.81 $\pm$ 0.35	93.75 $\pm$ 0.45	82.02 $\pm$ 0.49	41.89 $\pm$ 5.83	80.36 $\pm$ 1.06	54.04 $\pm$ 1.34
GraphMAE2	29.24 $\pm$ 1.16	84.97 $\pm$ 0.74	90.59 $\pm$ 0.79	51.48 $\pm$ 0.59	68.96 $\pm$ 2.44	40.81 $\pm$ 3.37	71.57 $\pm$ 1.15	91.53 $\pm$ 0.32	93.62 $\pm$ 0.54	82.22 $\pm$ 0.58	47.57 $\pm$ 6.19	81.45 $\pm$ 0.76	55.05 $\pm$ 1.19
MaskGAE	29.94 $\pm$ 0.61	84.44 $\pm$ 1.09	90.38 $\pm$ 0.67	50.13 $\pm$ 0.71	70.03 $\pm$ 0.82	43.84 $\pm$ 3.90	68.66 $\pm$ 1.40	90.11 $\pm$ 0.50	92.31 $\pm$ 0.49	80.67 $\pm$ 0.78	45.95 $\pm$ 4.52	81.95 $\pm$ 0.92	55.23 $\pm$ 0.99
Node4All	31.93 $\pm$ 1.79	81.88 $\pm$ 0.67	91.04 $\pm$ 0.59	42.09 $\pm$ 0.79	77.68 $\pm$ 1.02	42.99 $\pm$ 4.82	71.45 $\pm$ 0.58	91.72 $\pm$ 0.47	93.90 $\pm$ 0.45	80.57 $\pm$ 0.59	62.97 $\pm$ 5.80	76.15 $\pm$ 0.86	56.42 $\pm$ 1.87

Methods	Mine-sweeper	PubMed	Questions	Roman Empire	Squirrel	Texas	Tolokers	Wiki	Wisconsin	WikiCS	Full-Cora	OGBN Arxiv	Total Avg. (25)
Fixed Hyperparameter Setting													
Linear	80.00 $\pm$ 0.00	68.57 $\pm$ 0.31	97.13 $\pm$ 0.03	63.69 $\pm$ 0.61	29.72 $\pm$ 1.79	73.63 $\pm$ 5.47	78.16 $\pm$ 0.00	68.23 $\pm$ 1.97	82.55 $\pm$ 2.83	72.75 $\pm$ 0.53	34.15 $\pm$ 0.28	52.44 $\pm$ 0.03	62.79 $\pm$ 1.43
MLP	80.00 $\pm$ 0.00	70.57 $\pm$ 0.73	97.14 $\pm$ 0.13	65.12 $\pm$ 0.65	37.02 $\pm$ 2.58	75.52 $\pm$ 6.52	78.21 $\pm$ 0.20	75.57 $\pm$ 0.98	82.94 $\pm$ 6.01	71.32 $\pm$ 0.63	43.00 $\pm$ 0.56	54.60 $\pm$ 0.19	65.63 $\pm$ 1.64
GIN	74.10 $\pm$ 18.04	69.26 $\pm$ 8.89	87.44 $\pm$ 28.15	51.24 $\pm$ 1.29	32.64 $\pm$ 5.22	60.56 $\pm$ 5.80	55.63 $\pm$ 27.59	17.28 $\pm$ 1.75	49.41 $\pm$ 11.73	29.35 $\pm$ 5.55	57.86 $\pm$ 0.57	60.12 $\pm$ 4.91	50.75 $\pm$ 7.58
GAT	81.27 $\pm$ 0.76	75.73 $\pm$ 0.92	97.07 $\pm$ 0.03	43.16 $\pm$ 0.79	33.74 $\pm$ 1.29	58.42 $\pm$ 6.01	80.03 $\pm$ 0.46	43.67 $\pm$ 5.12	53.33 $\pm$ 5.93	78.78 $\pm$ 0.65	59.80 $\pm$ 0.61	71.51 $\pm$ 0.21	64.58 $\pm$ 2.09
GAT2	81.61 $\pm$ 0.75	75.12 $\pm$ 0.99	97.07 $\pm$ 0.03	45.50 $\pm$ 1.14	34.02 $\pm$ 1.86	55.16 $\pm$ 6.41	81.13 $\pm$ 0.36	30.97 $\pm$ 6.23	50.20 $\pm$ 4.22	79.03 $\pm$ 0.65	59.03 $\pm$ 0.80	72.12 $\pm$ 0.19	64.04 $\pm$ 2.17
SGC	80.14 $\pm$ 0.11	76.65 $\pm$ 0.13	97.11 $\pm$ 0.02	44.26 $\pm$ 0.57	38.18 $\pm$ 1.57	57.60 $\pm$ 4.18	78.68 $\pm$ 0.38	71.18 $\pm$ 1.03	52.16 $\pm$ 4.31	78.71 $\pm$ 0.55	59.49 $\pm$ 0.56	68.97 $\pm$ 0.05	66.36 $\pm$ 1.47
S ² GC	80.18 $\pm$ 0.17	76.54 $\pm$ 0.11	97.09 $\pm$ 0.01	46.96 $\pm$ 0.54	36.58 $\pm$ 1.11	57.88 $\pm$ 4.14	78.68 $\pm$ 0.33	71.97 $\pm$ 1.27	47.65 $\pm$ 5.34	79.06 $\pm$ 0.73	60.17 $\pm$ 0.53	69.16 $\pm$ 0.01	66.48 $\pm$ 1.38
JKNet	80.39 $\pm$ 0.32	76.56 $\pm$ 0.42	97.07 $\pm$ 0.03	53.87 $\pm$ 0.86	34.23 $\pm$ 1.15	58.42 $\pm$ 4.91	80.66 $\pm$ 0.44	72.35 $\pm$ 1.02	53.53 $\pm$ 3.40	78.45 $\pm$ 0.36	63.05 $\pm$ 0.50	71.81 $\pm$ 0.33	67.78 $\pm$ 1.38
APPNP	80.04 $\pm$ 0.10	76.90 $\pm$ 0.16	97.13 $\pm$ 0.05	56.14 $\pm$ 0.54	33.67 $\pm$ 1.30	55.95 $\pm$ 4.29	78.10 $\pm$ 0.19	71.19 $\pm$ 1.40	49.80 $\pm$ 4.57	78.66 $\pm$ 0.67	63.56 $\pm$ 0.38	70.76 $\pm$ 0.16	67.97 $\pm$ 1.34
GCN	80.13 $\pm$ 0.17	76.18 $\pm$ 0.27	97.14 $\pm$ 0.03	45.35 $\pm$ 0.40	33.64 $\pm$ 0.86	58.96 $\pm$ 5.23	79.45 $\pm$ 0.58	70.25 $\pm$ 0.71	52.35 $\pm$ 4.30	78.55 $\pm$ 0.74	62.82 $\pm$ 0.43	70.88 $\pm$ 0.28	66.88 $\pm$ 1.34
GCN2	86.10 $\pm$ 0.56	75.75 $\pm$ 1.22	97.20 $\pm$ 0.03	75.81 $\pm$ 0.76	34.96 $\pm$ 1.84	56.79 $\pm$ 5.79	82.11 $\pm$ 0.83	75.33 $\pm$ 1.55	61.37 $\pm$ 5.26	78.78 $\pm$ 0.81	60.04 $\pm$ 0.44	71.89 $\pm$ 0.31	69.76 $\pm$ 1.63
SAGE	86.30 $\pm$ 0.63	75.91 $\pm$ 0.24	97.23 $\pm$ 0.06	77.18 $\pm$ 0.65	37.46 $\pm$ 1.26	66.82 $\pm$ 8.62	80.28 $\pm$ 0.89	73.94 $\pm$ 1.37	67.45 $\pm$ 4.57	78.67 $\pm$ 0.52	61.49 $\pm$ 0.42	70.88 $\pm$ 0.12	70.81 $\pm$ 1.55
GPR-GNN	84.02 $\pm$ 0.47	76.64 $\pm$ 0.34	97.27 $\pm$ 0.04	73.33 $\pm$ 0.63	34.66 $\pm$ 1.42	68.75 $\pm$ 5.95	78.28 $\pm$ 0.13	72.30 $\pm$ 1.09	71.18 $\pm$ 4.73	79.14 $\pm$ 0.72	63.37 $\pm$ 0.49	68.39 $\pm$ 0.20	71.13 $\pm$ 1.43
UniMP	86.16 $\pm$ 0.52	76.31 $\pm$ 0.29	97.23 $\pm$ 0.04	79.26 $\pm$ 0.62	37.55 $\pm$ 1.35	67.93 $\pm$ 6.97	81.19 $\pm$ 0.65	74.70 $\pm$ 1.50	71.57 $\pm$ 4.04	79.26 $\pm$ 0.76	60.38 $\pm$ 0.54	70.83 $\pm$ 0.23	70.97 $\pm$ 1.48
node2vec	79.98 $\pm$ 0.04	65.27 $\pm$ 0.77	97.15 $\pm$ 0.05	62.91 $\pm$ 0.38	38.57 $\pm$ 2.25	74.73 $\pm$ 5.29	81.00 $\pm$ 0.54	79.77 $\pm$ 0.73	80.98 $\pm$ 5.48	78.29 $\pm$ 0.46	56.34 $\pm$ 0.36	70.10 $\pm$ 0.07	69.69 $\pm$ 1.42
GAE	80.11 $\pm$ 0.28	77.67 $\pm$ 0.63	97.12 $\pm$ 0.02	45.86 $\pm$ 0.41	41.33 $\pm$ 1.50	53.54 $\pm$ 6.56	81.06 $\pm$ 0.55	68.78 $\pm$ 1.53	50.20 $\pm$ 2.51	78.79 $\pm$ 0.88	63.31 $\pm$ 0.56	72.02 $\pm$ 0.08	67.22 $\pm$ 1.34
BGRL	80.24 $\pm$ 0.30	78.37 $\pm$ 1.12	97.15 $\pm$ 0.04	51.45 $\pm$ 0.46	41.84 $\pm$ 1.11	53.51 $\pm$ 3.97	80.95 $\pm$ 0.80	72.06 $\pm$ 1.52	52.94 $\pm$ 4.64	78.24 $\pm$ 0.62	62.39 $\pm$ 0.58	71.45 $\pm$ 0.18	68.00 $\pm$ 1.34
CCA-SSG	80.13 $\pm$ 0.17	79.37 $\pm$ 0.27	97.13 $\pm$ 0.03	54.80 $\pm$ 0.50	42.82 $\pm$ 2.38	56.50 $\pm$ 6.12	80.81 $\pm$ 0.50	71.71 $\pm$ 0.79	50.98 $\pm$ 3.82	78.45 $\pm$ 0.76	62.31 $\pm$ 0.40	71.66 $\pm$ 0.22	68.83 $\pm$ 1.46
GraphMAE	80.10 $\pm$ 0.14	78.63 $\pm$ 1.19	97.08 $\pm$ 0.04	46.68 $\pm$ 0.57	34.80<								

Table 2: Performance (%) across 25 node classification benchmarks comparing 14 supervised methods and 7 graph representation learning methods. All baselines use *per-dataset hyperparameter tuning* (Tuned-HP), with hyperparameters selected based on validation accuracy for each dataset. The **red and **second-best** methods in terms of average accuracy are highlighted. Despite not using any dataset-specific training or hyperparameter optimization, Node4All ranks 5th out of 21 methods, demonstrating competitive performance against extensively tuned, dataset-specific baselines.**

Methods	Actor	Amazon Computer	Amazon Photo	Amazon Ratings	Blog Catalog	Chameleon	Citeseer	Coauthor CS	Coauthor Physics	Cora	Cornell	DBLP	Deezer
Per-dataset Hyperparameter Tuning													
Linear	34.35 $\pm$ 0.96	69.75 $\pm$ 1.29	77.77 $\pm$ 1.17	38.61 $\pm$ 0.44	68.97 $\pm$ 1.34	33.52 $\pm$ 2.73	49.05 $\pm$ 0.74	84.35 $\pm$ 0.50	84.24 $\pm$ 1.78	47.53 $\pm$ 0.51	69.73 $\pm$ 7.33	55.28 $\pm$ 0.76	55.04 $\pm$ 2.39
MLP	34.82 $\pm$ 0.95	71.00 $\pm$ 1.07	78.55 $\pm$ 1.42	46.14 $\pm$ 0.47	73.81 $\pm$ 1.14	39.95 $\pm$ 3.21	51.40 $\pm$ 1.52	87.01 $\pm$ 0.62	88.65 $\pm$ 0.82	55.73 $\pm$ 0.80	71.62 $\pm$ 6.65	55.72 $\pm$ 0.66	57.23 $\pm$ 1.42
GIN	26.53 $\pm$ 1.29	35.26 $\pm$ 6.71	25.79 $\pm$ 3.29	44.93 $\pm$ 3.34	18.21 $\pm$ 2.49	33.28 $\pm$ 2.54	53.56 $\pm$ 4.91	84.97 $\pm$ 1.02	79.19 $\pm$ 18.57	54.41 $\pm$ 20.55	42.43 $\pm$ 8.38	65.07 $\pm$ 9.35	55.40 $\pm$ 0.39
GAT	28.55 $\pm$ 0.92	85.73 $\pm$ 0.62	90.99 $\pm$ 0.62	49.08 $\pm$ 0.94	48.47 $\pm$ 9.79	37.32 $\pm$ 2.39	68.10 $\pm$ 0.62	89.85 $\pm$ 0.62	91.89 $\pm$ 0.82	80.79 $\pm$ 0.92	41.62 $\pm$ 5.82	76.74 $\pm$ 2.47	55.40 $\pm$ 0.58
GAT2	28.57 $\pm$ 1.28	85.30 $\pm$ 0.79	90.34 $\pm$ 0.66	49.76 $\pm$ 0.41	57.70 $\pm$ 3.96	37.55 $\pm$ 4.65	67.65 $\pm$ 1.12	89.38 $\pm$ 1.03	91.59 $\pm$ 1.12	79.55 $\pm$ 1.64	41.62 $\pm$ 7.07	75.38 $\pm$ 3.28	55.80 $\pm$ 0.29
SGC	29.05 $\pm$ 1.09	85.40 $\pm$ 1.06	91.79 $\pm$ 0.46	42.41 $\pm$ 0.64	71.38 $\pm$ 1.04	41.70 $\pm$ 2.22	67.29 $\pm$ 0.46	90.17 $\pm$ 0.55	92.83 $\pm$ 0.48	78.46 $\pm$ 0.25	39.73 $\pm$ 7.36	73.33 $\pm$ 3.09	54.90 $\pm$ 1.61
S ² GC	29.63 $\pm$ 1.06	85.77 $\pm$ 0.95	91.86 $\pm$ 0.65	42.15 $\pm$ 0.38	78.74 $\pm$ 1.40	38.79 $\pm$ 3.01	66.71 $\pm$ 0.32	90.81 $\pm$ 0.55	91.74 $\pm$ 0.90	78.21 $\pm$ 0.28	38.11 $\pm$ 8.59	69.98 $\pm$ 3.33	54.48 $\pm$ 1.49
JKNet	29.95 $\pm$ 1.03	85.88 $\pm$ 1.02	91.66 $\pm$ 0.56	48.76 $\pm$ 0.79	71.48 $\pm$ 0.98	40.88 $\pm$ 3.81	69.38 $\pm$ 0.71	90.55 $\pm$ 0.56	93.33 $\pm$ 0.49	81.04 $\pm$ 0.85	45.14 $\pm$ 5.14	73.89 $\pm$ 4.01	55.97 $\pm$ 0.66
APPNP	31.63 $\pm$ 0.80	85.69 $\pm$ 1.12	92.35 $\pm$ 0.38	50.33 $\pm$ 0.54	85.43 $\pm$ 1.60	38.35 $\pm$ 2.38	68.79 $\pm$ 0.57	91.48 $\pm$ 0.60	93.30 $\pm$ 0.53	81.50 $\pm$ 0.54	49.19 $\pm$ 5.90	75.32 $\pm$ 3.14	56.46 $\pm$ 1.52
GCN	29.76 $\pm$ 0.73	85.63 $\pm$ 0.94	91.73 $\pm$ 0.49	49.49 $\pm$ 0.46	70.82 $\pm$ 1.01	39.89 $\pm$ 2.92	68.67 $\pm$ 0.69	90.83 $\pm$ 0.56	93.02 $\pm$ 0.46	81.58 $\pm$ 0.67	42.16 $\pm$ 5.43	75.07 $\pm$ 3.48	55.46 $\pm$ 1.60
GCN2	33.83 $\pm$ 0.76	85.26 $\pm$ 1.10	91.69 $\pm$ 0.73	49.95 $\pm$ 0.54	86.39 $\pm$ 2.30	39.35 $\pm$ 3.06	67.95 $\pm$ 0.77	91.31 $\pm$ 0.51	92.96 $\pm$ 0.59	80.50 $\pm$ 0.53	41.89 $\pm$ 6.42	75.16 $\pm$ 2.59	56.57 $\pm$ 1.22
SAGE	35.66 $\pm$ 0.91	83.81 $\pm$ 0.69	91.41 $\pm$ 0.57	50.54 $\pm$ 0.55	80.17 $\pm$ 2.77	40.40 $\pm$ 3.47	67.40 $\pm$ 1.36	90.54 $\pm$ 0.53	92.90 $\pm$ 0.46	80.14 $\pm$ 0.76	59.46 $\pm$ 15.81	72.05 $\pm$ 2.38	56.49 $\pm$ 1.25
GPR-GNN	34.91 $\pm$ 0.54	86.47 $\pm$ 0.99	93.19 $\pm$ 0.47	48.81 $\pm$ 0.64	88.15 $\pm$ 1.84	38.65 $\pm$ 3.21	69.96 $\pm$ 0.97	92.01 $\pm$ 0.67	93.33 $\pm$ 0.65	82.22 $\pm$ 1.01	59.46 $\pm$ 7.35	76.03 $\pm$ 2.93	56.40 $\pm$ 1.10
UniMP	35.52 $\pm$ 0.93	84.91 $\pm$ 0.79	92.02 $\pm$ 0.72	50.37 $\pm$ 0.48	76.58 $\pm$ 3.25	39.26 $\pm$ 4.37	67.00 $\pm$ 0.42	90.39 $\pm$ 0.72	92.38 $\pm$ 0.63	80.11 $\pm$ 1.15	62.43 $\pm$ 4.75	71.85 $\pm$ 2.69	56.75 $\pm$ 1.28
node2vec	34.31 $\pm$ 0.87	79.47 $\pm$ 0.75	87.64 $\pm$ 1.27	50.98 $\pm$ 0.31	79.81 $\pm$ 1.35	40.55 $\pm$ 2.54	61.76 $\pm$ 0.55	88.96 $\pm$ 0.87	89.96 $\pm$ 1.26	73.73 $\pm$ 0.86	70.27 $\pm$ 4.68	70.70 $\pm$ 0.84	55.60 $\pm$ 1.97
GAE	30.09 $\pm$ 0.56	85.63 $\pm$ 0.83	91.52 $\pm$ 0.48	50.80 $\pm$ 0.46	73.63 $\pm$ 1.28	43.48 $\pm$ 5.24	65.27 $\pm$ 1.50	91.42 $\pm$ 0.46	93.70 $\pm$ 0.23	81.75 $\pm$ 0.80	24.05 $\pm$ 6.22	80.70 $\pm$ 1.11	55.45 $\pm$ 1.63
BGRL	30.22 $\pm$ 1.06	84.90 $\pm$ 0.58	91.47 $\pm$ 0.58	51.01 $\pm$ 0.61	74.51 $\pm$ 1.16	44.69 $\pm$ 2.21	64.35 $\pm$ 0.82	91.65 $\pm$ 0.39	93.79 $\pm$ 0.51	81.81 $\pm$ 0.42	45.95 $\pm$ 6.04	80.45 $\pm$ 1.19	55.50 $\pm$ 1.40
CCA-SSG	36.48 $\pm$ 0.69	85.99 $\pm$ 0.85	91.31 $\pm$ 0.67	51.39 $\pm$ 0.34	85.05 $\pm$ 0.67	45.87 $\pm$ 2.55	68.89 $\pm$ 0.51	91.65 $\pm$ 0.23	94.03 $\pm$ 0.32	82.37 $\pm$ 0.33	74.59 $\pm$ 5.16	82.12 $\pm$ 0.68	59.65 $\pm$ 1.95
GraphMAE	31.35 $\pm$ 0.92	86.68 $\pm$ 0.96	91.73 $\pm$ 0.46	51.94 $\pm$ 0.60	69.88 $\pm$ 1.82	41.87 $\pm$ 2.86	70.48 $\pm$ 1.09	92.23 $\pm$ 0.42	93.75 $\pm$ 0.43	81.69 $\pm$ 0.28	45.99 $\pm$ 6.04	81.65 $\pm$ 0.39	55.68 $\pm$ 0.79
GraphMAE2	29.03 $\pm$ 0.87	86.58 $\pm$ 0.93	92.03 $\pm$ 0.57	51.85 $\pm$ 0.31	69.08 $\pm$ 1.98	34.16 $\pm$ 3.76	71.59 $\pm$ 0.30	92.11 $\pm$ 0.28	93.90 $\pm$ 0.48	82.76 $\pm$ 0.40	47.30 $\pm$ 5.95	81.96 $\pm$ 0.57	55.60 $\pm$ 0.82
MaskGAE	30.10 $\pm$ 0.81	86.04 $\pm$ 0.98	91.13 $\pm$ 0.83	50.65 $\pm$ 0.57	72.49 $\pm$ 0.75	43.39 $\pm$ 2.99	70.76 $\pm$ 0.72	91.15 $\pm$ 0.45	93.51 $\pm$ 0.33	81.29 $\pm$ 0.57	53.24 $\pm$ 7.36	82.61 $\pm$ 0.91	55.54 $\pm$ 0.78
Node4All	31.93 $\pm$ 1.79	81.88 $\pm$ 0.67	91.04 $\pm$ 0.59	42.09 $\pm$ 0.79	77.68 $\pm$ 1.02	42.99 $\pm$ 4.82	71.45 $\pm$ 0.58	91.72 $\pm$ 0.47	93.90 $\pm$ 0.45	80.57 $\pm$ 0.59	62.97 $\pm$ 5.80	76.15 $\pm$ 0.86	56.42 $\pm$ 1.87

Methods	Mine-sweeper	PubMed	Questions	Roman Empire	Squirrel	Texas	Tolokers	Wiki	Wisconsin	WikiCS	Full-Cora	OGBN Arxiv	Total Avg. (25)
Per-dataset Hyperparameter Tuning													
Linear	80.00 $\pm$ 0.00	68.57 $\pm$ 0.31	97.13 $\pm$ 0.03	63.69 $\pm$ 0.61	29.72 $\pm$ 1.79	73.63 $\pm$ 5.47	78.15 $\pm$ 0.02	72.06 $\pm$ 0.73	82.55 $\pm$ 2.83	72.75 $\pm$ 0.53	34.15 $\pm$ 0.28	52.44 $\pm$ 0.03	62.92 $\pm$ 1.38
MLP	80.00 $\pm$ 0.00	70.57 $\pm$ 0.73	97.27 $\pm$ 0.05	65.42 $\pm$ 0.58	38.06 $\pm$ 1.46	75.52 $\pm$ 6.52	78.01 $\pm$ 0.42	75.45 $\pm$ 0.83	83.14 $\pm$ 5.96	72.57 $\pm$ 0.65	46.47 $\pm$ 0.24	53.80 $\pm$ 0.14	65.92 $\pm$ 1.53
GIN	80.15 $\pm$ 0.22	69.07 $\pm$ 9.64	97.03 $\pm$ 0.04	52.43 $\pm$ 1.19	35.05 $\pm$ 1.22	60.35 $\pm$ 5.62	78.16 $\pm$ 0.00	33.27 $\pm$ 18.37	49.61 $\pm$ 11.60	59.28 $\pm$ 12.64	58.11 $\pm$ 0.49	65.01 $\pm$ 1.95	54.26 $\pm$ 5.83
GAT	81.52 $\pm$ 1.06	75.75 $\pm$ 1.28	97.09 $\pm$ 0.03	43.37 $\pm$ 0.39	34.90 $\pm$ 1.68	58.69 $\pm$ 4.36	81.22 $\pm$ 0.47	62.90 $\pm$ 1.87	54.12 $\pm$ 5.70	78.80 $\pm$ 0.53	60.50 $\pm$ 0.83	71.81 $\pm$ 0.20	65.81 $\pm$ 1.82
GAT2	82.87 $\pm$ 0.88	76.83 $\pm$ 0.36	97.06 $\pm$ 0.04	60.82 $\pm$ 0.79	34.83 $\pm$ 1.94	57.03 $\pm$ 8.00	81.19 $\pm$ 0.73	56.36 $\pm$ 1.94	55.49 $\pm$ 7.89	79.01 $\pm$ 0.77	60.84 $\pm$ 0.47	71.98 $\pm$ 0.14	66.58 $\pm$ 2.05
SGC	81.40 $\pm$ 0.49	77.47 $\pm$ 0.09	97.11 $\pm$ 0.02	44.26 $\pm$ 0.57	38.47 $\pm$ 2.20	57.60 $\pm$ 4.18	78.67 $\pm$ 0.25	71.66 $\pm$ 1.09	48.43 $\pm$ 6.74	78.71 $\pm$ 0.55	60.06 $\pm$ 0.64	68.69 $\pm$ 0.02	66.44 $\pm$ 1.49
S ² GC	81.04 $\pm$ 0.46	76.54 $\pm$ 1.11	97.11 $\pm$ 0.02	48.53 $\pm$ 0.38	36.54 $\pm$ 1.24	59.21 $\pm$ 5.08	78.73 $\pm$ 0.27	72.61 $\pm$ 1.14	48.63 $\pm$ 5.10	78.97 $\pm$ 0.58	59.86 $\pm$ 0.45	69.16 $\pm$ 0.01	66.56 $\pm$ 1.51
JKNet	81.36 $\pm$ 0.46	76.25 $\pm$ 1.29	97.11 $\pm$ 0.05	54.58 $\pm$ 3.72	34.66 $\pm$ 1.33	57.61 $\pm$ 6.84	80.46 $\pm$ 0.75	72.26 $\pm$ 1.08	52.75 $\pm$ 4.76	79.03 $\pm$ 0.71	62.91 $\pm$ 0.39	71.84 $\pm$ 0.24	67.95 $\pm$ 1.69
APPNP	80.94 $\pm$ 0.33	76.79 $\pm$ 0.16	97.12 $\pm$ 0.05	60.87 $\pm$ 0.36	33.92 $\pm$ 1.37	57.06 $\pm$ 5.18	78.15 $\pm$ 0.22	74.72 $\pm$ 0.56	53.53 $\pm$ 6.45	78.90 $\pm$ 0.52	63.80 $\pm$ 0.45	70.73 $\pm$ 0.15	69.05 $\pm$ 1.42
GCN	80.61 $\pm$ 0.49	77.66 $\pm$ 0.75	97.11 $\pm$ 0.06	45.44 $\pm$ 0.45	36.24 $\pm$ 1.75	60.04 $\pm$ 6.31	80.33 $\pm$ 0.58	70.99 $\pm$ 1.18	54.12 $\pm$ 4.31	78.76 $\pm$ 0.66	63.00 $\pm$ 0.57	71.22 $\pm$ 0.22	67.59 $\pm$ 1.47
GCN2	86.48 $\pm$ 0.60	76.24 $\pm$ 0.54	97.20 $\pm$ 0.06	76.09 $\pm$ 0.61	34.12 $\pm$ 0.99	59.23 $\pm$ 4.89	81.86 $\pm$ 0.91	76.45 $\pm$ 1.20	62.35 $\pm$ 5.39	78.59 $\pm$ 0.75	61.30 $\pm$ 0.53	72.04 $\pm$ 0.13	70.19 $\pm$ 1.51
SAGE	87.03 $\pm$ 0.77	76.41 $\pm$ 0.92	97.24 $\pm$ 0.06	78.22 $\pm$ 0.62	38.84 $\pm$ 1.74	74.15 $\pm$ 8.66	80.55 $\pm$ 0.77	73.95 $\pm$ 1.38	77.06 $\pm$ 5.89	78.42 $\pm$ 0.60	60.74 $\pm$ 0.48	71.06 $\pm$ 0.17	71.79 $\pm$ 2.14
GPR-GNN	84.28 $\pm$ 0.37	77.15 $\pm$ 0.40	97.27 $\pm$ 0.05	73.61 $\pm$ 0.58	35.24 $\pm$ 1.71	68.76 $\pm$ 6.63	78.24 $\pm$ 0.12	75.52 $\pm$ 0.51	76.08 $\pm$ 5.93	79.51 $\pm$ 0.68	64.36 $\pm$ 0.42	69.34 $\pm$ 0.28	71.96 $\pm$ 1.60
UniMP	87.90 $\pm$ 0.43	75.41 $\pm$ 0.85	97.24 $\pm$ 0.05	80.54 $\pm$ 0.77	37.21 $\pm$ 1.96	69.82 $\pm$ 8.66	81.85 $\pm$ 0.60	75.02 $\pm$ 1.01	77.45 $\pm$ 2.52	79.13 $\pm$ 0.52	59.46 $\pm$ 0.50	71.70 $\pm$ 0.19	71.69 $\pm$ 1.61
node2vec	79.91 $\pm$ 0.16	68.33 $\pm$ 1.59	97.15 $\pm$ 0.05	62.60 $\pm$ 0.49	37.96 $\pm$ 2.49	73.62 $\pm$ 4.83	81.05 $\pm$ 0.65	79.71 $\pm$ 0.70	82.55 $\pm$ 3.45	78.17 $\pm$ 0.65	56.34 $\pm$ 0.36	70.02 $\pm$ 0.11	70.05 $\pm$ 1.35
GAE	80.41 $\pm$ 0.33	78.83 $\pm$ 0.86	91.74 $\pm$ 0.90	49.98 $\pm$ 0.53	43.87 $\pm$ 2.10	64.39 $\pm$ 5.87	80.96 $\pm$ 0.48	68.63 $\pm$ 1.35	54.12 $\pm$ 5.76	79.23 $\pm$ 0.91	63.96 $\pm$ 0.39	72.02 $\pm$ 0.08	67.82 $\pm$ 1.61
BGRL	80.24 $\pm$ 0.26	79.64 $\pm$ 0.37	97.16 $\pm$ 0.04	51.75 $\pm$ 0.28	41.37 $\pm$ 1.49	59.77 $\pm$ 6.00	81.10 $\pm$ 0.69	72.05 $\pm$ 0.90	59.80 $\pm$ 5.63	79.01 $\pm$ 0.59	63.45 $\pm$ 0.31	71.47 $\pm$ 0.15	69.08 $\pm$ 1.35
CCA-SSG	80.11 $\pm$ 0.23	79.42 $\pm$ 0.40	97.30 $\pm$ 0.03	65.44 $\pm$ 0.66	43.08 $\pm$ 1.75	75.53 $\pm$ 4.81	81.30 $\pm$ 0.47	78.16 $\pm$ 0.87	81.96 $\pm$ 4.71	79.32 $\pm$ 0.54	63.73 $\pm$ 0.46	72.04 $\pm$ 0.24	73.87 $\pm$ 1.21
GraphMAE	80.41 $\pm$ 0.18	80.41 $\pm$ 0.43	97.11 $\pm$ 0.01	49.84 $\pm$ 0.38	37.19 $\pm$ 1.65	65.22 $\pm$ 3.97	78.						